

Step 1 — Import libraries

```
In [1]: import pandas as pd

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

Step 2 — Read the CSV file

```
In [17]: df = pd.read_csv("bank_cleaned.csv")

df.head()
```

```
Out[17]:
```

	age	job	marital	education	default	balance	housing	loan	contact	day	mc
0	58	4	1	2	0	1.839343	1	0	2	-1.288418	
1	44	9	2	1	0	-0.696791	1	0	2	-1.288418	
2	33	2	1	1	0	-0.729183	1	1	2	-1.288418	
3	47	1	1	3	0	1.075143	1	0	2	-1.288418	
4	33	11	2	3	0	-0.730382	0	0	2	-1.288418	

Step 3 — Check the Target Variable

```
In [18]: print(df["y"].value_counts())
```

```
y
0    26679
1     1659
Name: count, dtype: int64
```

Step 4 — Select Features

```
In [19]: X = df[["age", "balance", "duration", "campaign"]]
```

Step 5 — Select Target

```
In [20]: y = df["y"]
```

Step 6 —Split the Dataset

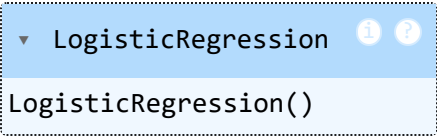
```
In [21]: X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

Step 7 —Create the Logistic Regression Model

```
In [22]: model = LogisticRegression()
```

Step 8 — Train the Model

```
In [23]: model.fit(X_train, y_train)
```

```
Out[23]:  LogisticRegression()  
LogisticRegression()
```

Step 9: Make Predictions

```
In [25]: y_pred = model.predict(X_test)  
y_pred
```

```
Out[25]: array([0, 0, 0, ..., 0, 0, 0])
```

Step 10: Calculate Accuracy

```
In [26]: accuracy = accuracy_score(y_test, y_pred)  
  
print("Accuracy:", accuracy)
```

```
Accuracy: 0.9393083980239944
```

```
In [ ]:
```